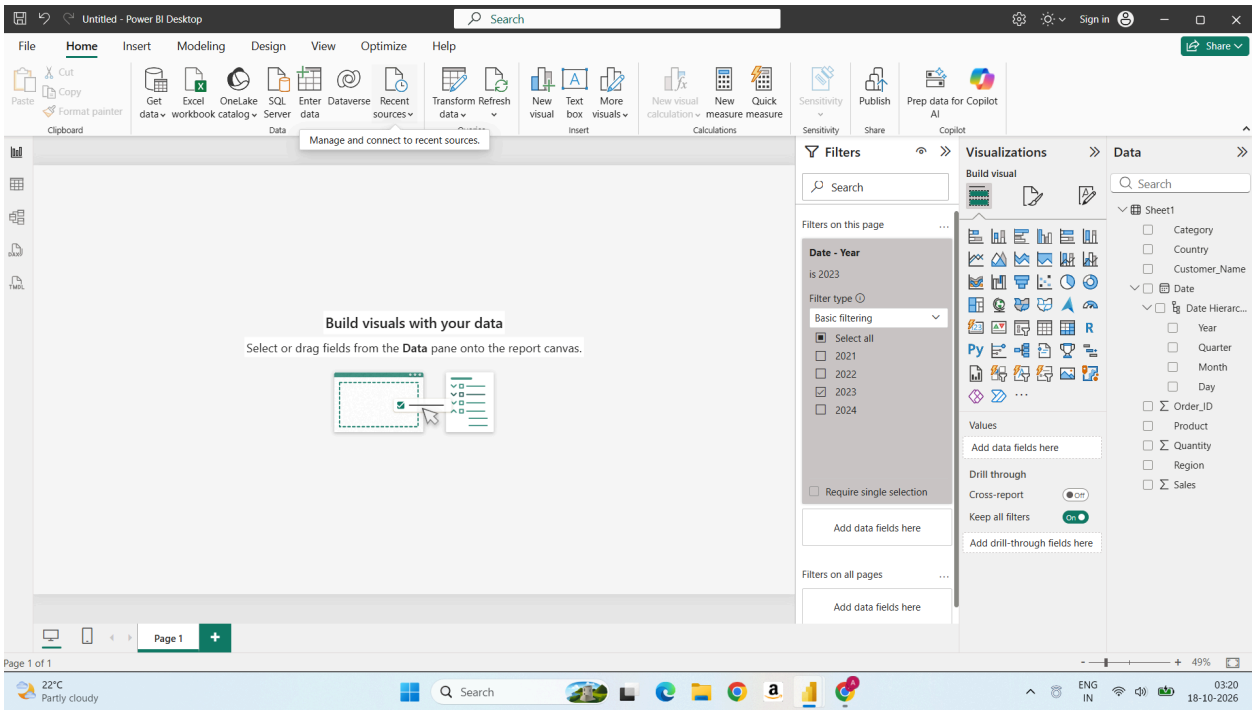
Advanced Visuals in PowerBI

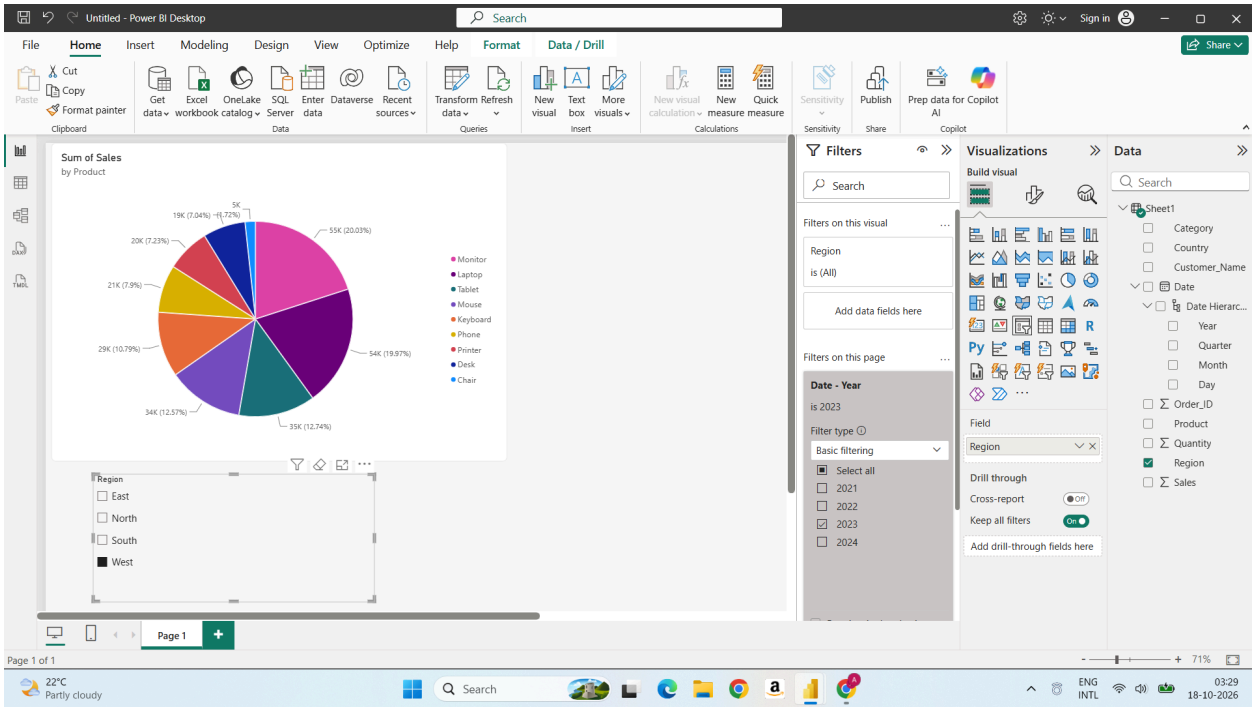
1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

- **Basic Filter:** Allows users to check or uncheck specific individual values from a list of items present in a selected field (e.g., selecting only "2023" under Year).
  PNG
- **Advanced Filter:** Allows complex logical rules and conditions (e.g., *Contains*, *Starts with*, *Is greater than*, *Is less than*) combined with AND / OR operators to filter text or numeric data dynamically.
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- **Slicer:** An on-canvas visual filter element that enables end-users to interactively filter report pages directly from the report view without needing to open the Filters pane.
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2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.



3. Create a Pie Chart of Sales by Product and add a Slicer that allows users to filter the dataset by Region.



4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

This is a key concept question in Power BI. Here is the simple, clear answer:

1. Visual-Level Filter

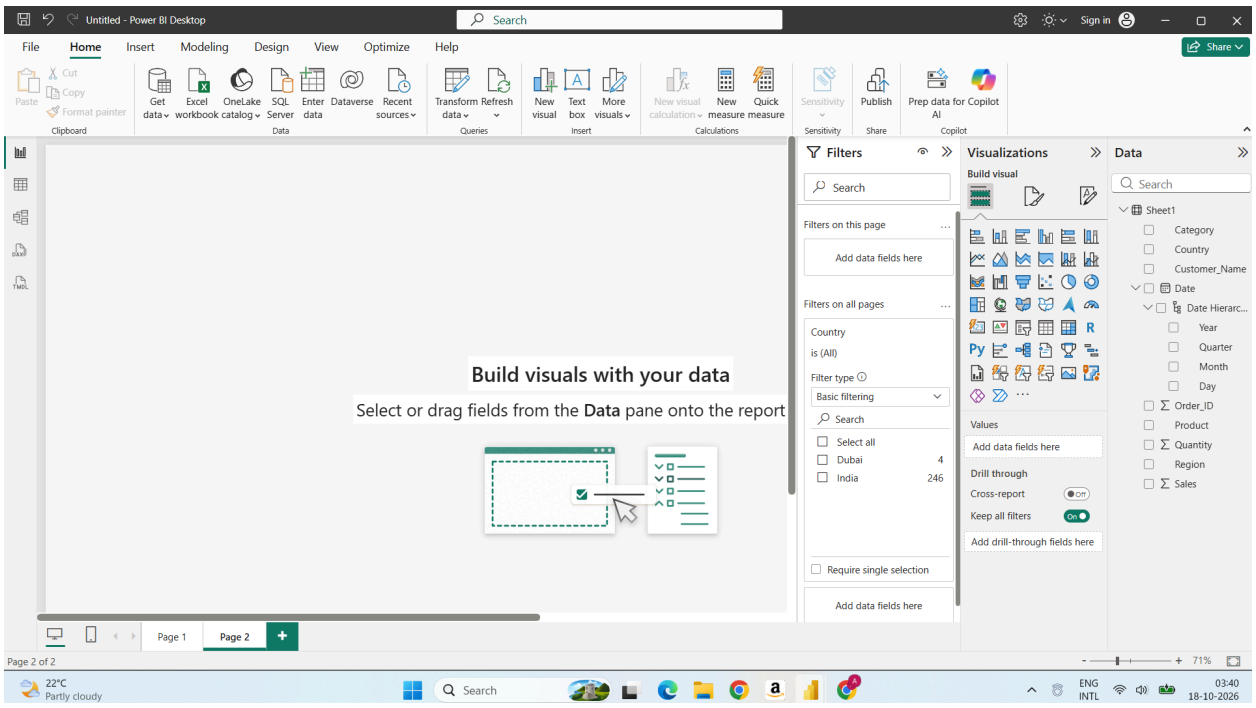
- **What it does:** Applies a filter to **only one specific chart or table** that you select on your canvas.
- **Scope:** It affects *only* that single visual and leaves all other charts on the page untouched.

2. Difference between the 3 Filter Levels:

Filter Scope	What it affects	Example

Visual-Level	**1 visual only**	Filtering a single Bar Chart to show only *Laptops*, while other charts on the page show all products.
Page-Level	**All visuals on the current page**	Filtering Page 1 to show only *2023* data, so every chart on Page 1 shows 2023.
Report-Level	**All visuals across all pages**	Filtering the entire file to show only *India*, so Page 1, Page 2, and Page 3 all only show India data.

5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.



6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

What is the role of Drill-Down?

Drill-Down allows users to navigate from a high-level summary view down into more detailed levels of data within the same visual using a set hierarchy.

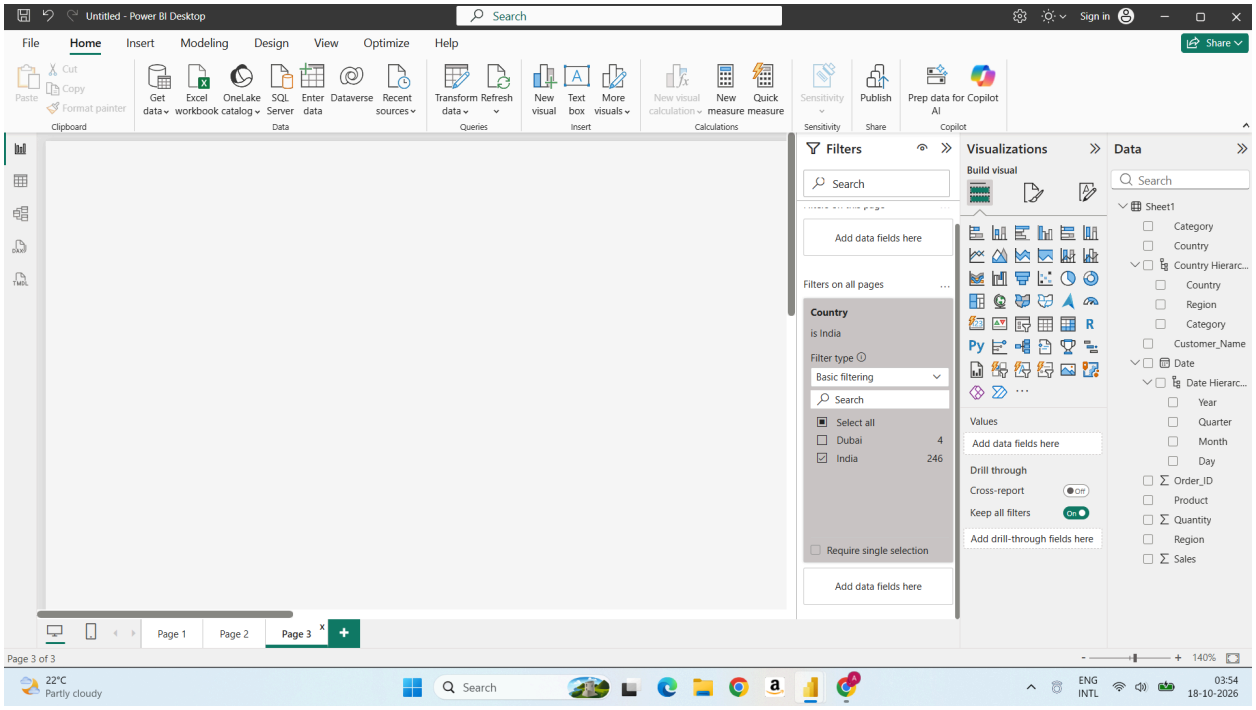
It lets you "zoom in" on a specific category to see its sub-components without cluttering the report with multiple charts.

- **Example Hierarchy:** Country → Region → Product Category
- **How it works:** Clicking on a specific country (e.g., *India*) drills down to display only the *Regions* inside India, and clicking on a region (e.g., *North*) drills down further into its *Product Categories*.

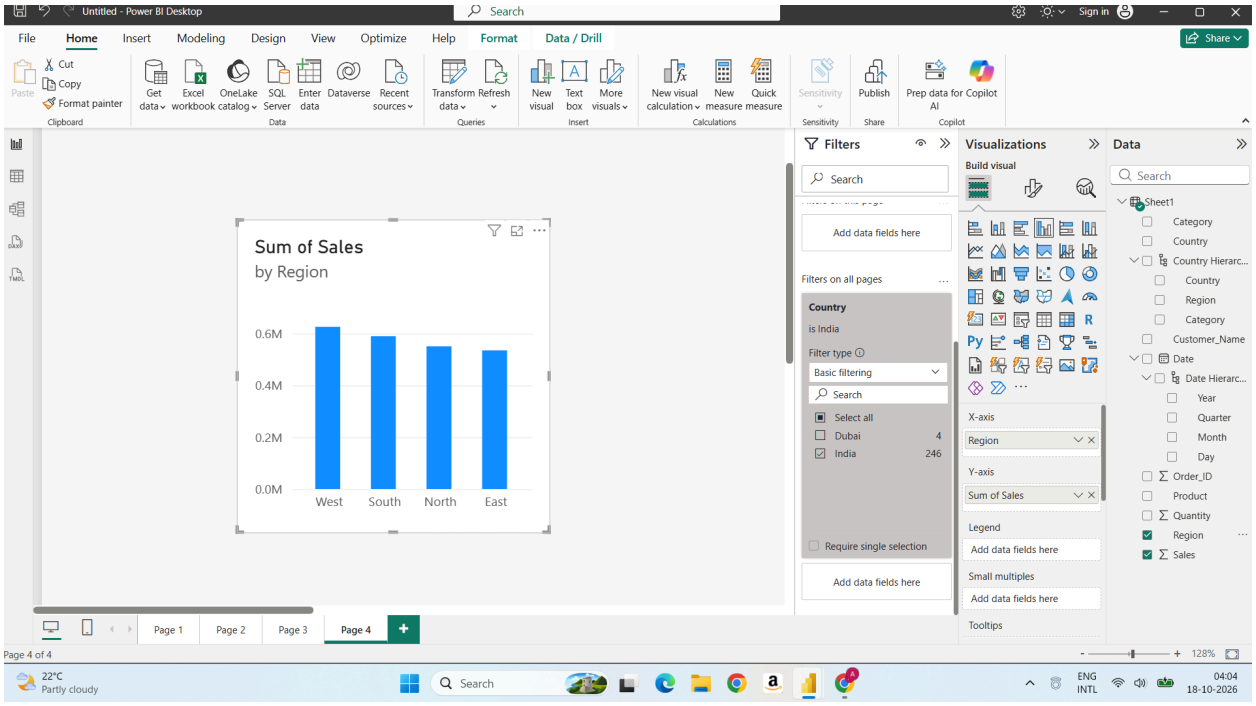
How does it improve data exploration?

1. **Reduces Dashboard Clutter:** Instead of creating separate charts for Countries, Regions, and Categories, you can combine them all into a single interactive chart.
2. **Enables Intuitive Root-Cause Analysis:** Users can easily spot an high-level anomaly (e.g., low overall sales) and quickly drill down layer by layer to find the exact region or product causing it.
3. **Interactive User Experience:** It gives end-users freedom to explore data at their desired level of detail dynamically.

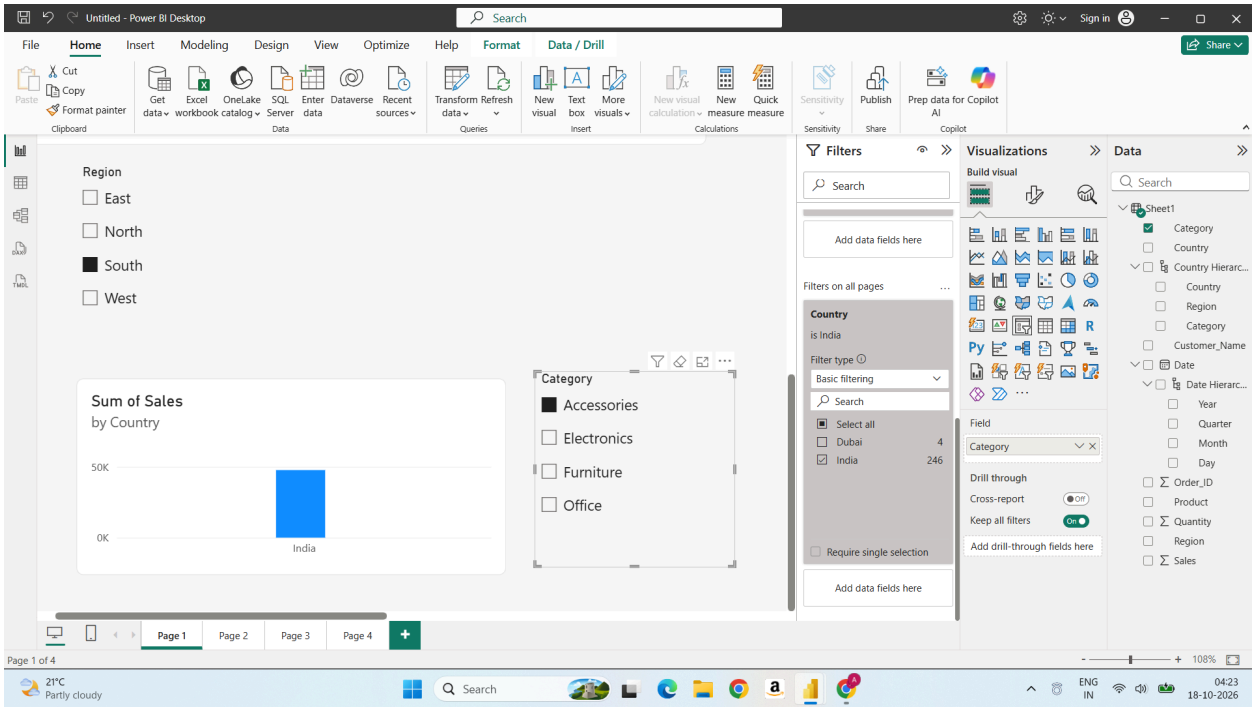
7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.



8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region-wise sales.



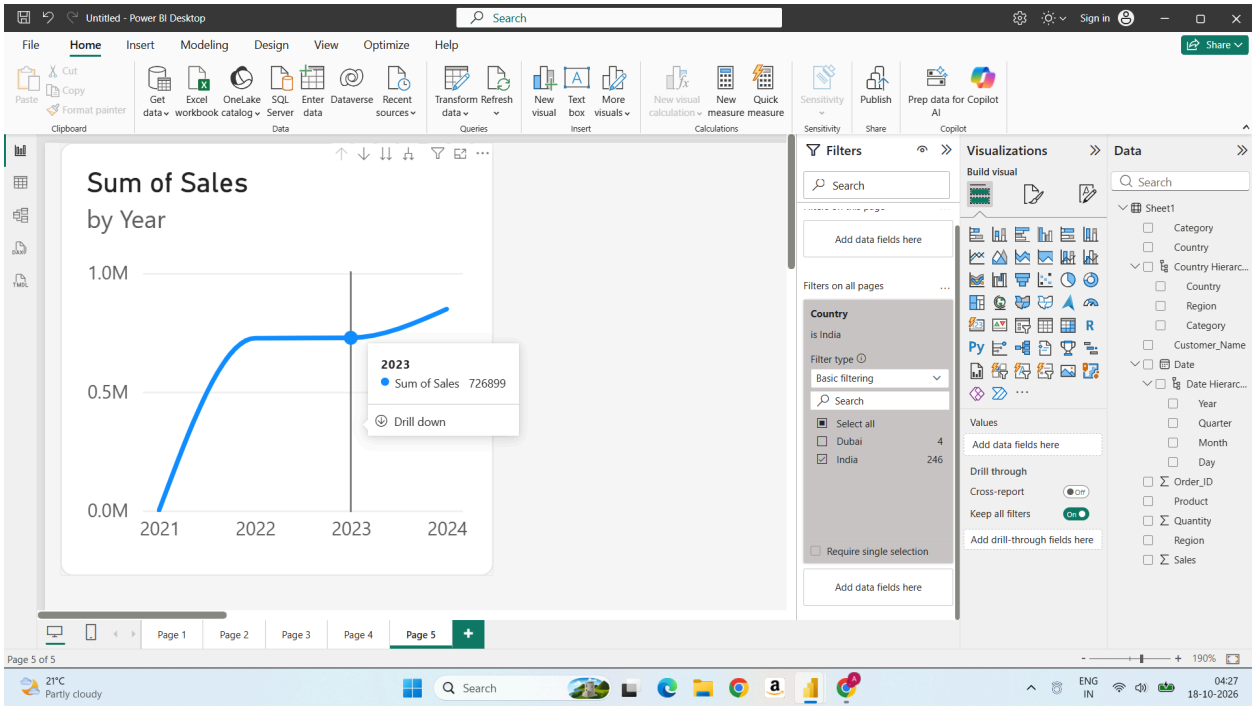
9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

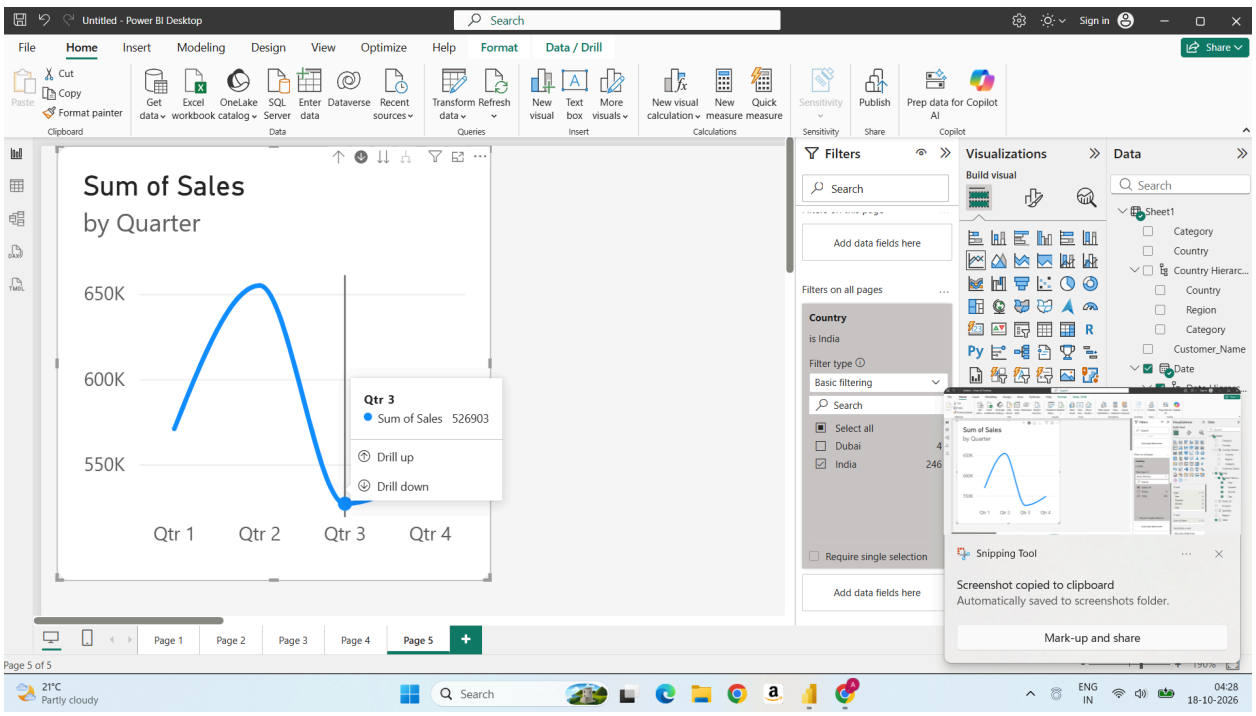


10. Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

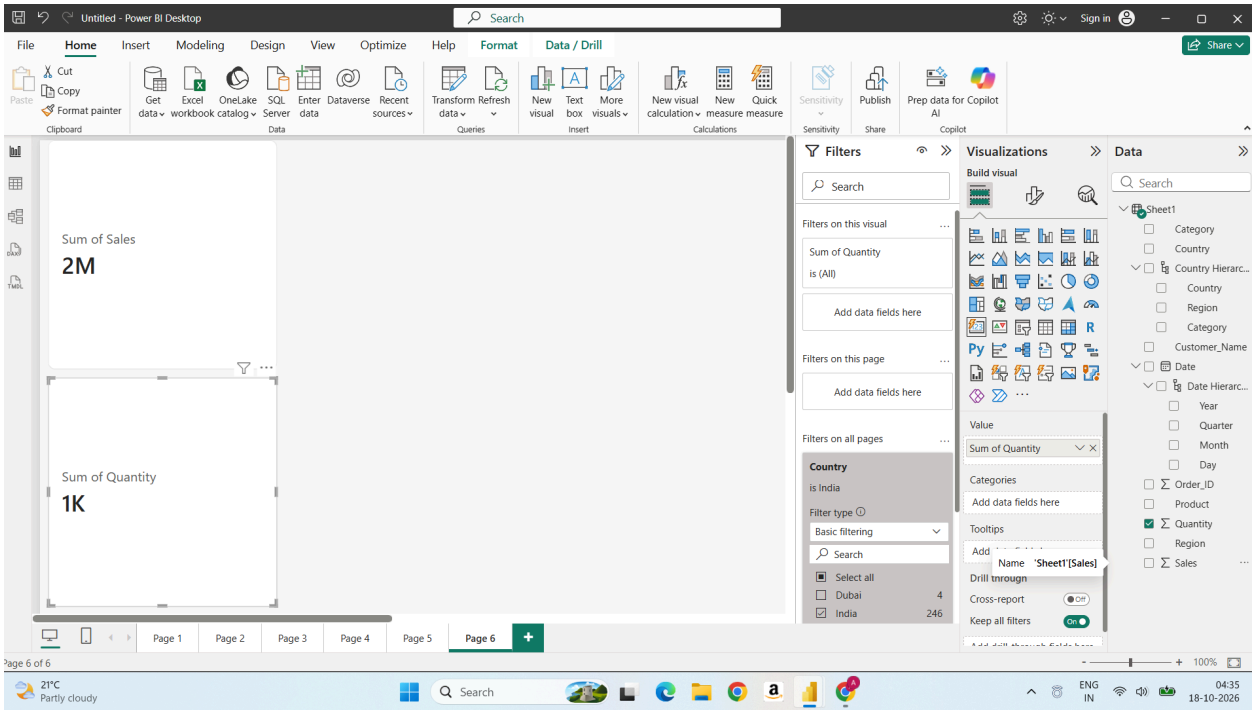
- **Drill-Down:** Navigates **deeper into hierarchical levels of detail within the exact same visual**.
  PNG+ 1
  - *Real-world Example:* Clicking on the "2023" column in a yearly sales chart to instantly see quarters (Q1, Q2, Q3, Q4) displayed on that same chart.
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- **Drill-Through:** Navigates **from a summary visual to an entirely separate, dedicated detailed report page** based on a selected context item.
  PNG+ 1
  - *Real-world Example:* Right-clicking on the product "Laptop" in a summary dashboard and clicking "Drill-Through" to open a separate **Laptop Deep-Dive Page** showing regional breakdown, customer names, and order quantities.
    PNG+ 1

11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.





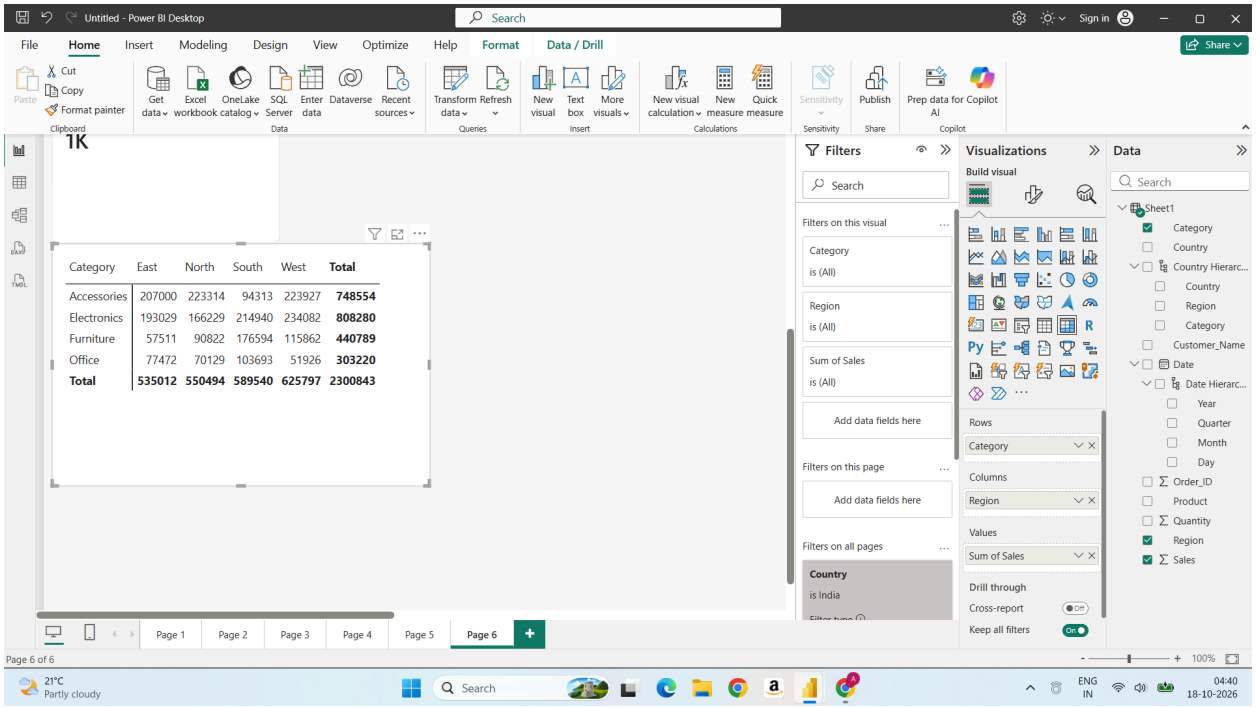
12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000.



13. What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction?

- **Definition:** "Keep All Filters" is a configuration toggle switch in Power BI's drill-through setup panel on the destination page.
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- **Effect on User Interaction:**
  - **When ON:** All active slicers, page-level filters, and cross-filters applied on the source page (e.g., Region = North, Year = 2023) are carried over and applied to the destination drill-through page.
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  - **When OFF:** Only the specific clicked item's context (e.g., the specific product selected) is passed to the target page, ignoring any other active slicers or page filters from the source page.
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14. Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.



15. Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data.